
THE EFFECTS OF JOB MATCHING ON EMPLOYMENT ACCESSIBILITY MEASURES IN THE ASSESSMENT OF CORE PUBLIC TRANSPORT INFRASTRUCTURE

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J Rafael Verduzco-Torres *

Urban Studies

University of Glasgow

Hillhead, Glasgow G12 8RZ, United Kingdom

JoseRafael.Verduzco-Torres@glasgow.ac.uk

David Philip McArthur

Urban Studies

University of Glasgow

Hillhead, Glasgow G12 8RZ, United Kingdom

David.McArthur@glasgow.ac.uk

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Abstract

The definition of accessibility emphasizes the attractiveness of opportunities that residents deem valuable. This study revises the common assumption in empirical studies that residents are equally attracted by all type of employment and its implication for public transport evaluation. Additionally, the role of the modifiable problem unit is also examined in this relationship.

Keywords Employment accessibility · Job matching · Public transport infrastructure · Modifiable areal unit problem

1 Introduction

Location-based accessibility measures are key to account for the level of service offered by the transport infrastructure in a variety of studies, including the property market or equity outcomes of large public transport infrastructure projects. An elemental definition of accessibility emphasizes the attractiveness of opportunities that residents deem valuable. This study revises the common assumption in empirical studies that residents are equally attracted by all type of employment. Additionally, the role of the modifiable problem unit is also examined in this relationship. These considerations can offer useful guidance for the

*Corresponding author. Email: JoseRafael.Verduzco-Torres@glasgow.ac.uk.

environmental measure of accessibility and its application in environmental studies and the distribution of the benefits of public transport infrastructure.

1. What are the key spatial differences between generic and matching accessibility measures and what is the role of the MAUP?
2. How does the evaluation of PT infrastructure differs when accounting for matching employment opportunities in accessibility measures?
3. How do benefits of PT extensions differ between deprived or low-skilled populations and the general population when considering employment matching opportunities in accessibility?

This paper contributes to literature by:

- Examining the spatial distribution of public transport benefits using accessibility measures by relaxing the assumption that all employment opportunities are equally attractive to all the population drawing on an empirical case in a ten-year period.
- Examines the role of the MAUP in this effects.
- Reports the findings of GMC, an understudied mega-city, by identifying and digitally reconstructing in a common format (general transit feed specification, GTFS) a series of temporal stages of the MPTN in GMC for a ten-year period (2009-2019).

2 Literature review

Thériault, Voisin, and Des Rosiers (2013) conceptualize accessibility as “the ability that persons living in that area can, within an acceptable travel time, reach a satisfactory diversity of activity locations that are important to them” (p. 234). Levinson and Wu (2020) simply put it as “the ease of reaching valued destinations” (p. 130).

2.1 Accessibility measures

formulation in Eq. ?? incorporates: (1) individual’s profiles; 2) how different profiles perceive or experience the space; and 3) constraints that may prevent segments of the population to access some type of activities. Some of these individual aspects involved in the construction of accessibility measures have been found to be relevant in empirical studies. For instance, gender in Osland (2010) or age and gender in Thériault, Voisin, and Des Rosiers (2013) in the study of the property prices.

Equation 1 shows the formulation potential accessibility to employment from location A_i , which the frequently-used in literature.

$$A_i = \sum_{j=1}^N E_j f(c_{ij}) \quad (1)$$

The attractiveness of a destination is given by the number of employment opportunities at j , E_j . $f(c_{ij})$ is the impedance affecting attractiveness as a function of travel cost between i and j . Frequently, travel cost is represented by the modelled travel time.

In Eq. ??, households and opportunities are assumed to be homogeneous, i.e., all types of employment are equally attractive for all types of residents. Following contributions from theory (Geurs and van Wee 2004; Levinson and Wu 2020), an extended accessibility measure should take into consideration the characteristics of the demand (residents at origins) and supply at potential destinations. In the context of the labour market, this idea is supported by the Human Capital Theory (Schultz 1961; Becker 1994).

There are different versions of its operationalization in existing empirical studies (e.g. Pereira, Grégoire, and Karner 2019). For instance, these characteristics can be expressed as type of person m at j and type of employment m at k . Accordingly, the measure can be specified as following:

$$A_{im} = \sum_{j=1}^N E_{im} f(c_{ij}), \quad (2)$$

Here, only employment opportunities E matching the profile of residents between j and k are considered. Thus, this reflect the notion that residents would find attractive only employment opportunities that match their skills and education level with a relevant employment opportunity.

2.1.1 Evaluation of public transport

Location-based accessibility measures have been used to evaluate the performance of public transport systems in policy report and research (Ermagun 2021; Liu, Porr, and Miller 2022; Conway, Byrd, and Van Eggermond 2018; Malekzadeh and Chung 2020; Palmateer et al. 2021). A measure based on realistic travel times can reflect important elements of a public transport system. This is because travel time by public transport is affected by several aspects of the system, such as:

- The topology of the network, e.g. connectivity, centrality.
- Capacity of the infrastructure, e.g. degree of right of way segregation, capacity and design of stations.
- Spatial coverage, e.g. availability of infrastructure and catchment areas.
- Operational characteristics, e.g. frequency of services, overlapping routes, express services, transfer wait times.

Despite the growing relevance of accessibility measures in the evaluation of public transport infrastructure, knowledge of the implications of overlooking matching employment are limited in literature.

3 Methods

3.1 Accessibility measures examined

The general accessibility is computed as following for this work:

$$A_{jt}^{\text{PT}} = \sum_{j=1}^J E_k f(d_{jkt}) \quad (3)$$

The implementation of matching accessibility is adapted according to the information available for the area of study and some preliminary findings. The households' education level (i.e., median number of years of education) is employed assuming correspondence with the level of salary paid in the employment market. This is further supported by the positive relationship found between education level and wages in empirical studies in the context of Mexico (Nuñez, Paredes, and Garduño-Rivera 2017; Rojas, Angulo, and Velázquez 2000).

$$A_{jtm}^{\text{PT}} = \sum_{j=1}^J E_{km} f(d_{jkt}), \quad (4)$$

Here, m at j is categorized according to the median education level of its residents by quintile and m at k is classified based on the average paid salary by quintile. Only employment opportunities E matching the education level quantile between j and k are considered (when $j \neq k$).² This assumes that residents would deem valuable only employment opportunities that match their education level, i.e. those in the same quintile. The data sources and specific procedures to process the data are described in the next sub-section.

3.2 Area of study

Greater Mexico City (GMC) (*Zona Metropolitana del Valle de México*) is located in the south-centre of Mexico. In 2015, GMC had nearly 21 million inhabitants (INEGI 2015b). The official delimitation consists of 76 adjacent municipalities, which are part of three different states, namely: (1) *Ciudad de México* (16 municipalities); (2) *Hidalgo* (1 municipality), and (3) *México* (59 municipalities). The total GMC area is 7,889 km², and it is home to 17% of the national population.

3.2.1 The main public transport network (MPTN)

The focus of this work is on *rapid transit* and *semi-rapid transit* infrastructure, excluding specialized modes i.e. *aerial tramway*. The combination of public transport modes included in the analyses is referred to as the *main public transport network* (MPTN). Figure 1 shows the MPTN in GMC for the year 2019. This consists

²An exception occurs when $j = k$. In this case, employment opportunities in the origin are considered as available. This is to avoid the assumption implying that some zones have accessibility values equal to 0

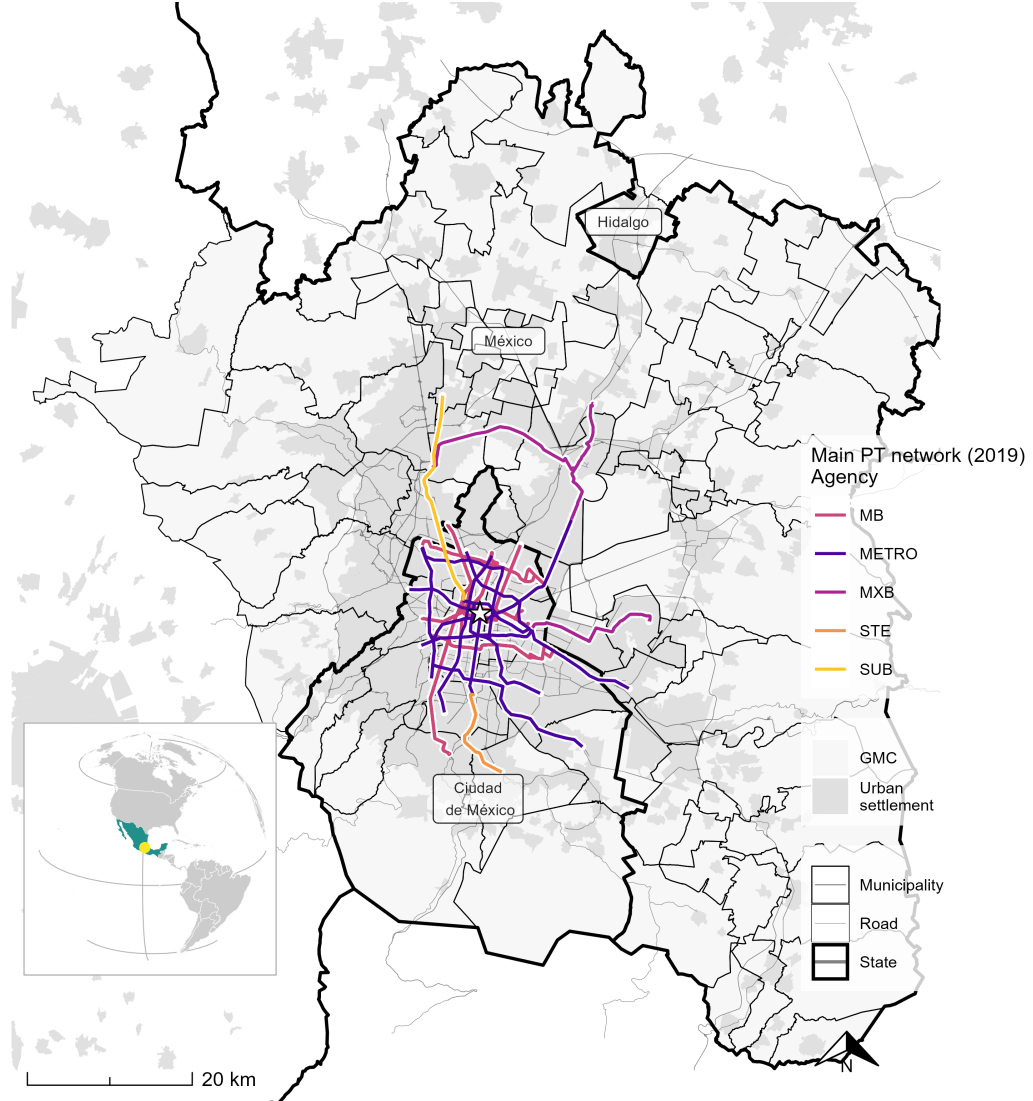


Figure 1: Main public transport network (MPTN). Greater Mexico City, 2019.

Source: Verduzco-Torres (2023) based on Marco Geoestadístico Nacional 2014 Versión 6.2 (INEGI, 2014); Delimitación de las zonas metropolitanas México 2015 (INEGI, 2018); Secretaría de Movilidad de la Ciudad de México (SEMOVI, n.d.); Sistema de Transporte Masivo y Teleférico del Estado de México (SITRAMyTEM, n.d.); OpenStreetMap (OSM, n.d.).

of a variety of modes operated by the following main agencies: The metro system, directly operated by the state-owned company *Sistema de Transporte Colectivo* (STC) and designated as Metro; two BRT systems, namely *Metrobus* (MB) in Ciudad de México and Mexibus (MXB) in the state of México; One LRT line operated by STE; and one regional rail line designated as *Sub*.

The period of study was divided into seven temporal stages to reflect significant changes in the MPTN over time. Initially, changes were identified by the physical alterations in the MPTN, such as the opening of a new corridor or the extension of an existing one. Table 1 details these changes and their corresponding temporal stages from 2010 to 2019. The stages were grouped based on the time interval between changes,

Table 1: Main public transport network extensions between 2010 and 2019 in Greater Mexico City grouped by temporal stage.

Corridor	Agency	From station	To station	Opening date	Previous change*	GTFS source
Stage 7: 2018 to 2019						
7	MB	Indio Verdes	Campo Marte	2018-03-01	26	2019-01-09
Stage 6: 2016 to 2018						
6	MB	El Rosario	Martín Carrera	2016-01-01	12	2017-09-02
Stage 5: 2015 to 2016						
2	MXB	La Quebrada	Las Americas	2015-01-01	14	2015-01-28
Stage 4: 2013 to 2015						
5	MB	Río de los Remedios	San Lázaro	2013-11-01	6	2014-12-30
3	MXB	Chimalhuacán	Pantitlán	2013-05-01	7	2014-12-30
Stage 3: 2012 to 2013						
12	Metro	Mixcoac	Tlahuac	2012-10-01	6	2013-10-21
4	MB	Buenavista	San Lazaro Aeropuerto	2012-04-01	14	2013-10-21
Stage 2: 2010 to 2012						
3	MB	Tenacayuca	Etiopia	2011-02-01	4	2013-10-21
1	MXB	Ciudad Azteca	Ojo de Agua	2010-10-01	10	2013-10-21
Stage 1: Before 2010						
2	MB	Tepalcates	Tacubaya	2009-12-01	NA	2013-10-21

Note:

*Months relative to the previous modification of the main public transport network.

with modifications occurring within six months being grouped together in the same stage. This criteria follows previous empirical studies and assumes that accessibility benefits are perceived as simultaneous when changes happen within short timeframes (Ahlfeldt 2013; Gibbons and Machin 2005).

The MPTN stages and corresponding corresponding corridors are in Figure 2.

3.3 Data

3.3.1 Demographics

The demographic data employed to account for person components in accessibility measures come from the 2010 Population Census (INEGI 2012). These data were selected for their detailed disaggregation at the city block level, including key fields such as total population and average years of education completed by individuals aged 15 and older.

The count data included were transferred to the SAS using the *areal weighting method* (Saporito et al. 2007). This reassigns the data according to the proportional overlay of the source and target areas. The average education level was assigned using the *population weighting method*. In addition, this takes into consideration the number of inhabitants and their type.

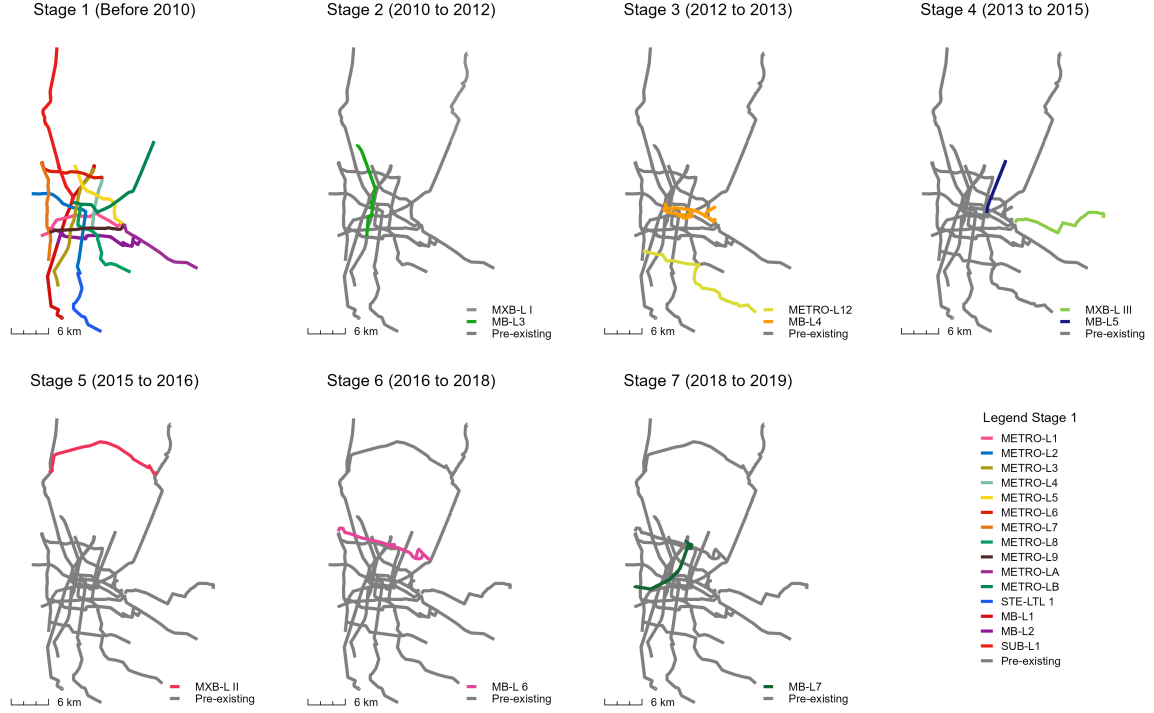


Figure 2: Temporal stages of the main public transport network. Greater Mexico City, 2010 to 2019.
 Source: The author based on Semovi (n.d.) and SITRAMyTEM (n.d.).

3.3.2 Employment

Employment data are sourced from the 2014 Economic Census (*Censos Económicos 2014*) (INEGI 2015b) and accessed at the city block level via the Microdata Lab and Remote Processing service (<https://www.inegi.org.mx/microdatos/>). The dataset includes: 1. Number of establishments (fixed locations producing goods or services); 2. Employed personnel (individuals active in establishments, including volunteers and family members), referred to as ‘employment’; and 3. Total gross remuneration paid for the relevant year.

Employment information which came aggregated at higher geographic level than the city-block due to privacy (<8%) were re-allocated using proportional estimates based on the spatial distribution from the 2015 DENU (INEGI 2015a). This is an open-access, point-level dataset of individual establishments. The employment data was assigned from city blocks to SASs following the same approach employed for the demographic data, i.e. according to the areal weighting method.

The average income was computed for each areal unit based on the total number of employment and the total gross remuneration paid.

3.3.3 Public transport itineraries

The estimated travel time used for accessibility measures is based on scheduled public transport services. The Transport Authority in Mexico City (*Secretaría de Movilidad de la Ciudad de México*, Semovi) released the

first *general transit feed specification* (GTFS) files in 2013, which were accessed via the Open Mobility website (<https://openmobilitydata.org/>). These datasets cover the agencies operating most MPTN services in GMC. However, there are some limitations: firstly, the geographic coverage primarily includes services within Ciudad de México, excluding BRT services provided by MXB. Secondly, the temporal coverage starts in 2013, with updates occurring only sporadically, approximately once a year.

To overcome the first limitation, the geographic coverage of the official GTFS data was augmented to include three BRT corridors operated by MXB. This was carried out *manually* based on unstructured information provided by the Transport Agency of the State of Mexico (*Sistema de Transporte Masivo y Teleferico del Estado de México*, Sitramytem). This corresponded to services for 2019 only.

Regarding the second limitation, the MPTN’s temporal stages were reconstructed using the GTFS standard. The primary input for each stage was the closest available official GTFS data reflecting changes in the MPTN, with only the relevant services for each corridor in their respective stages being retained. For MXB services, 2019 data was considered due to the lack of earlier details, assuming uniform characteristics of the BRT corridors across the stages. Similarly, for other corridors included in early stages, services were assumed to reflect 2013 levels due to the unavailability of earlier data. All non-MPTN services were excluded, thereby focusing exclusively on the benefits of the MPTN.

3.3.4 Road network and pedestrian infrastructure

The road and pedestrian infrastructure network is sourced from *OpenStreetMap* (OSM) (Manually downloaded on 30th October 2019 through a web export tool available in <https://export.hotosm.org/es/v3/>). This enables multi-modal journeys to realistically represent the access/egress stage to a public transport service when combining walking and public transport.

3.4 Travel costs and impedance parameters

3.4.1 Modeled travel time

Travel time matrices were calculated for each origin-destination (OD) pair across SASs using a combination of walking and public transport within the main public transport network. OD points i and j are represented by population-weighted centroids of spatial units containing economic activity (from the 2014 Economic Census) or demographic data (from the 2010 Population Census). Travel times are capped at 180 minutes, based on patterns observed in the 2017 Travel Survey.

Travel times were modelled using a door-to-door approach with OpenTripPlanner (OTP) version ‘1.4.0’ (OTP n.d.) via the implementation of an open-access routine written *Python* (Pereira et al. 2019). This approach considers access, waiting, onboard, transfer, and egress times. All-to-all travel time matrices were generated for each SAS across seven temporal stages using reconstructed GTFS files and road/pedestrian

networks, with eight random departure times between 10 a.m. and 2 p.m. The median travel time was used for accessibility models (Conway, Byrd, and van der Linden 2017). This process, computationally intensive due to the high resolution of SASs, resulted in 2.1 billion routes. Matrices were then merged, and median travel times calculated, with walking distances limited to 3 km when using public transport and 4.8 km (60 minutes) for walking-only trips.

The internal-zone travel-times were estimated based on the surface area as following: $d_{i=j} = \frac{1}{2} \sqrt{\frac{\text{area}_i}{\pi}}$ (Frost and Spence 1995; Stepniak and Rosik 2015). The metric distance was transformed to travel time using the average speed modelled for short journeys (less than 3 km).

3.4.2 Spatial Decaying Function and Calibration of Accessibility Parameters

The spatial decaying function and parameters for accessibility measures were determined using estimates from a trip distribution model, following Handy and Niemeier (1997). This model is used to understand how commuters in Greater Mexico City (GMC) discount employment opportunities based on travel time. The analyses use data from the 2017 Travel Survey, focusing on weekday public transport and walking trips.

A series of alternative specifications were estimated within a doubly constrained gravity model framework, expressed as following:

$$T_{jk} = A_j O_j B_k D_k f(d_{jk}) \quad (5)$$

where

$$A_j = \left[\sum B_i D_i f(d_{jk}) \right]^{-1} \quad (6)$$

and

$$B_k = \left[\sum A_k O_k f(d_{jk}) \right]^{-1} \quad (7)$$

with balancing factors A_j and B_k calculated iteratively. The model estimates commuter flows (T_{jk}) between origins (j) and destinations (k), with constraints on the number of journeys at each. $f(d_{jk})$ is a spatial deterrence or decaying function. The inverse power ($d_{jk}^{-\beta}$) and negative exponential ($e^{-\beta d_{jk}}$) decaying functions were tested. β represents the deterrence parameter.

A modified specification intended to capture an add-up cost for inter-zone journeys or analogously a benefit of residing and working within the same zone is considered (Thorsen and Gitlesen 1998). This includes a parameter μ which affects only the diagonal elements of the OD matrix. The term is given by the following expression:

$$\mu \delta_{jk} \quad (8)$$

where Kronecker delta takes the following value:

$$\delta_{ij} \begin{cases} 0 & \text{if } j \neq k \\ 1 & \text{if } j = k \end{cases} \quad (9)$$

The simultaneous calibration of the model parameters β and μ is achieved via optimization by minimizing the standardized root mean square error (SRMSE) using the *Nelder-Mead's simplex algorithm* as implemented in base R (**R-base?**).

The preliminary analyses shows that the negative exponential function provided more accurate estimates compared to the inverse power. Also, the use of the within-zone parameter μ improves the estimates. Therefore, the selected specification is the modified negative exponential including μ . The parameters employed to estimate accessibility are shown in Table .

4 Results

5 Conclusions

References

- Ahlfeldt, Gabriel M. 2013. “If We Build It, Will They Pay? Predicting Property Price Effects of Transport Innovations.” *Environment and Planning A* 45 (8, 8): 1977–94. <https://doi.org/10.1068/a4549>.
- Becker, Gary S. 1994. *Human Capital: A Theoretical and Empirical Analysis with Special Reference to Education*. 3rd ed. Chicago: The University of Chicago Press.
- Conway, Matthew Wigginton, Andrew Byrd, and Marco van der Linden. 2017. “Evidence-Based Transit and Land Use Sketch Planning Using Interactive Accessibility Methods on Combined Schedule and Headway-Based Networks.” *Transportation Research Record: Journal of the Transportation Research Board* 2653 (1, 1): 45–53. <https://doi.org/10.3141/2653-06>.
- Conway, Matthew Wigginton, Andrew Byrd, and Michael Van Eggermond. 2018. “Accounting for Uncertainty and Variation in Accessibility Metrics for Public Transport Sketch Planning.” *Journal of Transport and Land Use* 11 (1, 1). <https://doi.org/10.5198/jtlu.2018.1074>.
- Ermagun, Alireza. 2021. “Transit Access and Urban Space-Time Structure of American Cities.” *Journal of Transport Geography* 93 (May). <https://doi.org/10.1016/j.jtrangeo.2021.103066>.
- Frost, M. E., and N. A. Spence. 1995. “The Rediscovery of Accessibility and Economic Potential: The Critical Issue of Self-Potential.” *Environment & Planning A* 27 (11, 11): 1833–48. <https://doi.org/10.1068/a271833>.

- Geurs, Karst T., and Bert van Wee. 2004. “Accessibility Evaluation of Land-Use and Transport Strategies: Review and Research Directions.” *Journal of Transport Geography* 12 (2, 2): 127–40. <https://doi.org/10.1016/j.jtrangeo.2003.10.005>.
- Gibbons, Stephen, and Stephen Machin. 2005. “Valuing Rail Access Using Transport Innovations.” *Journal of Urban Economics* 57 (1, 1): 148–69. <https://doi.org/10.1016/j.jue.2004.10.002>.
- Handy, Susan L, and Deb A Niemeier. 1997. “Measuring Accessibility: An Exploration of Issues and Alternatives.” *Environment and Planning A* 29 (7, 7): 1175–94. <https://doi.org/10.1068/a291175>.
- INEGI. 2012. “Sistema Para La Consulta de Información Censal (SCINCE).” Instituto Nacional de Estadística y Geografía. 2012. <https://www.inegi.org.mx/app/descarga/?ti=13&ag=01>.
- . 2015a. “Directorio Estadístico Nacional de Unidades Económicas 2015.” 2015. <https://www.inegi.org.mx/default.html>.
- . 2015b. “Encuesta Intercensal 2015.” 2015. <https://www.inegi.org.mx/programas/intercensal/2015/>.
- Levinson, David M, and Hao Wu. 2020. “Towards a General Theory of Access.” *Journal of Transport and Land Use* 13 (1, 1): 129–58. <https://doi.org/10.5198/jtlu.2020.1660>.
- Liu, Luyu, Adam Porr, and Harvey J. Miller. 2022. “Realizable Accessibility: Evaluating the Reliability of Public Transit Accessibility Using High-Resolution Real-Time Data.” *Journal of Geographical Systems*, May. <https://doi.org/10.1007/s10109-022-00382-w>.
- Malekzadeh, Ali, and Edward Chung. 2020. “A Review of Transit Accessibility Models: Challenges in Developing Transit Accessibility Models.” *International Journal of Sustainable Transportation* 14 (10, 10): 733–48. <https://doi.org/10.1080/15568318.2019.1625087>.
- Núñez, Hector M., Dusan Paredes, and Rafael Garduño-Rivera. 2017. “Is Crime in Mexico a Disamenity? Evidence from a Hedonic Valuation Approach.” *Annals of Regional Science* 59 (1, 1): 171–87. <https://doi.org/10.1007/s00168-017-0823-8>.
- Osland, Liv. 2010. “Spatial Variation in Job Accessibility and Gender: An Intraregional Analysis Using Hedonic House-Price Estimation.” *Environment and Planning A* 42 (9, 9): 2220–37. <https://doi.org/10.1068/a42422>.
- OTP. n.d. “OpenTripPlanner.” Accessed October 10, 2019. <https://www.opentripplanner.org/>.
- Palmateer, Chelsey, Andrew Owen, David M Levinson, and Alireza Ermagun. 2021. “The Role of Transit Service Area Definition for Access-based Evaluation.” In *Applications of Access*. Sidney: The University of Sidney.
- Pereira, Rafael H M, David Banister, Tim Schwanen, and Nate Wessel. 2019. “Distributional Effects of Transport Policies on Inequalities in Access to Opportunities in Rio de Janeiro.” *Journal of Transport and Land Use* 12 (1 SE -, 1 SE -). <https://doi.org/10.5198/jtlu.2019.1523>.

- Pereira, Rafael H M, Laurent Grégoire, and Alex Karner. 2019. “Tutorial with Reproducible Example to Estimate a Travel Time Matrix Using OpenTripPlanner and Python.” <https://doi.org/10.5281/zenodo.3242134>.
- Rojas, Mariano, Humberto Angulo, and Irene Velázquez. 2000. “Rentabilidad de La Inversión En Capital Humano En México.” *Economía Mexicana. Nueva Época* IX (2, 2): 113–42. <http://hdl.handle.net/11651/4177>.
- Saporito, Salvatore, Jana M Chavers, Laura C Nixon, and Megan R McQuiddy. 2007. “From Here to There: Methods of Allocating Data Between Census Geography and Socially Meaningful Areas.” *Social Science Research* 36 (3, 3): 897–920. <https://doi.org/10.1016/j.ssresearch.2006.05.004>.
- Schultz, Theodore W. 1961. “Investment in Human Capital.” *The American Economic Review* 51 (1, 1): 1–17. <http://www.jstor.org/stable/1818907>.
- Stepniak, Marcin, and Piotr Rosik. 2015. “The Impact of Data Aggregation on Potential Accessibility Values.” In *Geoinformatics for Intelligent Transportation*, edited by Igor Ivan, Itzhak Benenson, Bin Jiang, Jiří Horák, James Haworth, and Tomáš Inspektor, 227–40. Cham: Springer International Publishing. https://doi.org/10.1007/978-3-319-11463-7_16.
- Thériault, Marius, Marion Voisin, and François Des Rosiers. 2013. “Accessibility of Urban Services: Modeling Socio-spatial Differences and Their Impacts on Residential Values.” In *Modeling Urban Dynamics: Mobility, Accessibility and Real Estate Value*, edited by Marius Thériault and François Des Rosiers, 225–54. London: ISTE Ltd and John Wiley & Sons, Inc.
- Thorsen, Inge, and Jens Petter Gitlesen. 1998. “Empirical Evaluation of Alternative Model Specifications to Predict Commuting Flows.” *Journal of Regional Science* 38 (2, 2): 273–92. <https://doi.org/10.1111/1467-9787.00092>.